MICHIGAN STATE
UNIVERSITY

Welcome to PHY 321: Classical Mechanics

Prof. Danny Caballero (he/they; course instructor)
Kang Yu (GTA)
Mihir Naik (he/him; ULA)

About Me

- Lappan Phillips Professor of Physics and Computational Science

- Co-direct two research labs (in Physics & Computational Science Education)

- PhD in Physics from Georgia Tech in 2011
  Postdoc CU-Boulder 2011-2013

- Former custodian, copy shop manager, and high school physics teacher in Atlanta Public Schools

- Labor Organizer for Union of Tenure System Faculty

- Completing my 13th year at MSU.

How to Contact Me

- Open "office" Hours (15 minutes)
  - https://cal.com/dannycaballero/phy-321
  - Zoom (in person if requested)
- Office hours
  - TBD or by appointment!
- Response Hours: 5am-4pm, M-F
  - We all work different hours
- Response time: 1-2 business days
  - After 2 days, please send me another email/message!
- Email (caball14@msu.edu)

Getting to know you!

- Student Information Survey is on website (Getting Started) and on D2L
    - Name & Pronouns; Python Experience; Access needs
- Fill this short survey out by **11:59pm Friday, Jan 16th** (https://forms.cloud.microsoft/r/7Ar26hXDgm)
- Fill out this form (linked in survey) for office hours by **11:59pm Friday, Jan 16th** (https://crab.fit/phy-321-spring-2026-office-hours-poll-464860)

- **Why is learning about you important?**
    - We care about you as an individual.
    - We should know what you need/want from this course.
    - We learn better in a community that supports us.

Important Links (some login with MSU credentials)

- Course website: https://dannycab.github.io/phy321msu/intro.html
- D2L – https://d2l.msu.edu/d2l/home/2375819
- Gradescope – https://www.gradescope.com/courses/1217180 (you will need a scanning device such as a phone camera, computer/tablet camera, etc.)

- **Please check that you can access D2L and that you've been added to the course on Gradescope.**

  Contact Danny if you have any issues!

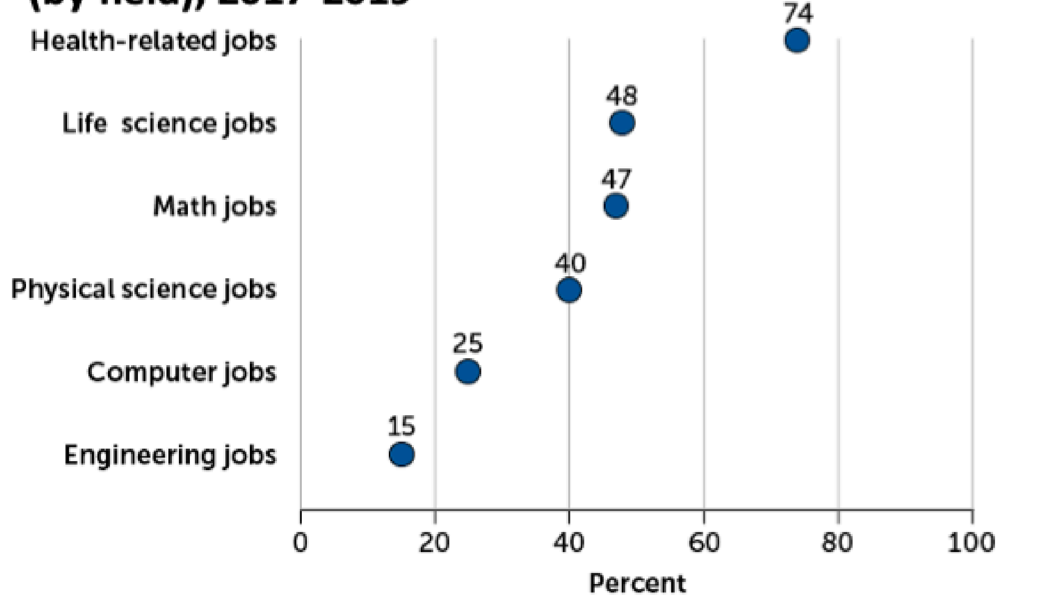
Course Description

- This course is designed to help you to learn classical physics concepts using both analytical and computational methods. Through examination of the fundamental principles of classical mechanics, we will gain a deeper understanding of how objects and systems move.
- Recommended Textbooks
  - (JRT): Taylor, J.R. (2005). *Classical mechanics.* University Science Books.
  - (AMS): Malthe-Sørenssen, A. (2015). *Elementary Mechanics Using Python: A Modern Course Combining Analytical and Numerical Techniques.* Springer.
- Supplementary Textbooks
  - (NLD): Strogatz, S. (2014). *Nonlinear Dynamics and Chaos.* CRC Press.
  - (MLB): Boas, M. L. (2006). *Mathematical methods in the physical sciences*. John Wiley & Sons.

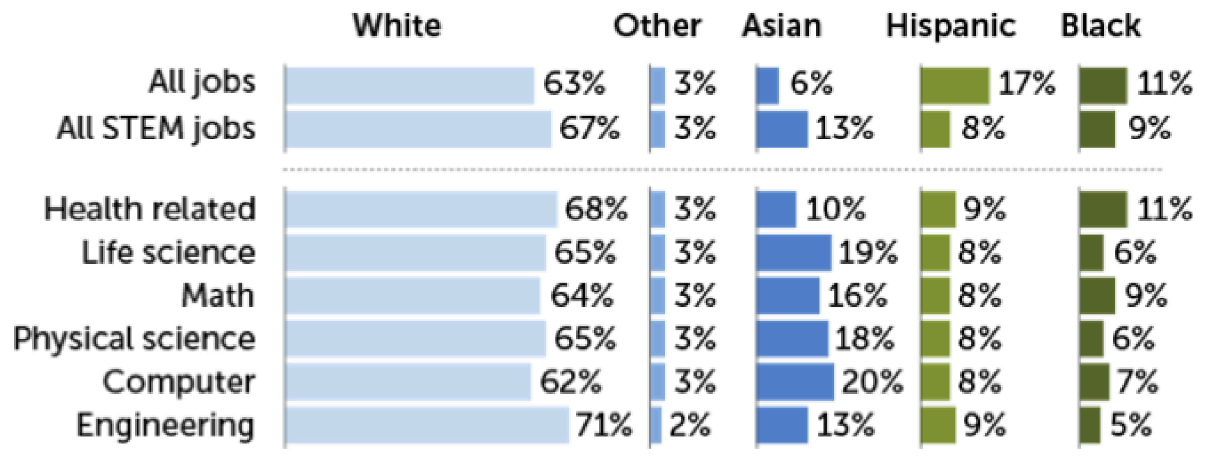
See "Textbooks" on course webpage for text access.

Racial and Gender Gaps Persist in STEM

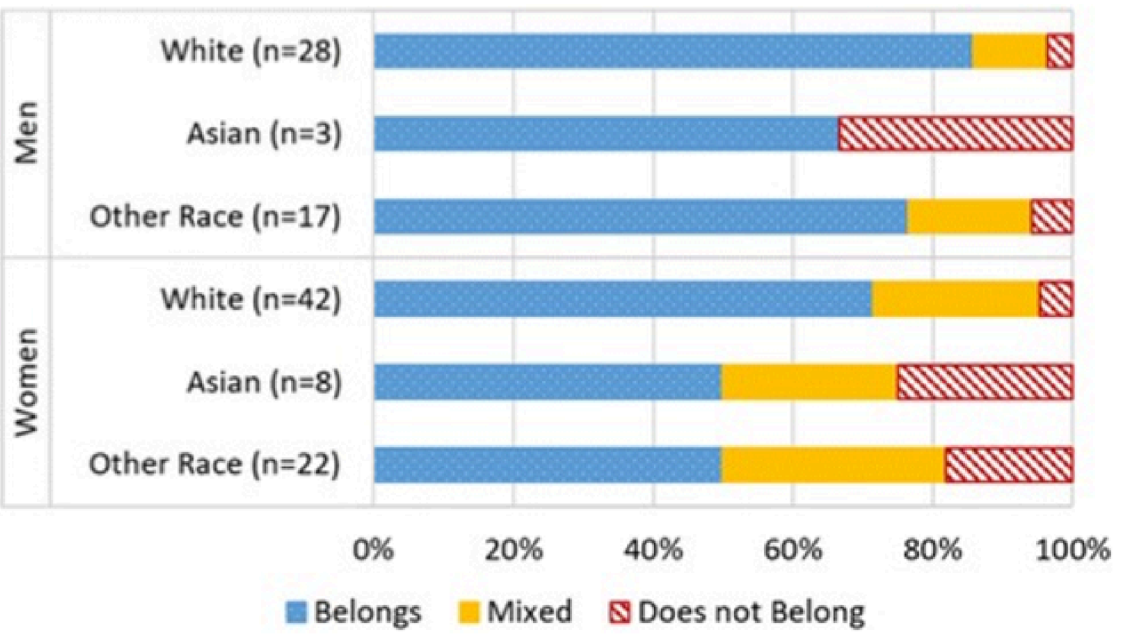
Percentage of STEM professionals who are women (by field), 2017-2019



Racial and Ethnic representation in STEM jobs (2017-2019)



Feelings of Belonging in STEM (by gender and race)



Figures taken from M. Temming, ScienceNews (2021) and K. Rainey et al., International Journal of STEM Education (2018)

Inclusive Classrooms

- **Inclusive classroom:** Where instructors and students work together to create and sustain an environment in which everyone feels safe, supported, and encouraged to express their views and concerns (UMich CTL definition)
- We strive to create inclusive classroom spaces, but we can't do it without you
- You will need to work together to solve the problems we present while making sure that everyone has the chance to contribute to the effort
- **Issues or concerns:** Please bring those to Danny. If you prefer, you can also contact our associate chair, Prof. Stuart Tessmer, or, if needed, the department chair, Prof. Steve Zepf.

Learning is a social and collaborative act!

- It's an essential skill in science and engineering (and highly valued by employers)!
- Social interactions are critical to scientists' success – most good ideas grow out of discussions with colleagues, and essentially all physicists work as part of a group.
- **For HW, you can work in groups (up to 3 people) OR by yourself. If you work as a group, you can turn in the assignment as a group – <u>REMEMBER TO WRITE YOUR NAME(S) AND ADD YOUR GROUP MEMBERS TO THE GRADESCOPE SUBMISSION!</u>**

Course Activities

Homework (15%) – 8 of them DUE Fridays at 11:59pm ET **via Gradescope**

- Workshop Days on Fridays
- Can work/submit in groups (up to 3 people) OR individually
- Each part graded on a 3-point rubric (correct, slightly incorrect, incorrect, blank)
- Will drop lowest

Homework is challenging and long; start early

Exercise 0 – See Reflection Activities

Course Activities

Reflection Activities (10%) – 3 forms of them **(turned in on D2L)**

1. Content checkins (due on the Sunday of each week at 11:59pm ET via the D2L Assignments page)

- What was the most challenging part of the pre-class materials, and how did you address it? This could be related to the content of the pre-class materials, your learning style, etc.
- What did you learn about your own strengths through engaging with the pre-class materials?
- What would you do differently if you were to approach the pre-class materials again and why?
- What questions do you have about the pre-class materials that you would like to discuss in class?

Course Activities

Reflection Activities (10%) – 3 forms of them **(turned in on D2L)**

2. Exercise 0 (due with HW via the D2L Assignments page)

The purpose of Exercise 0 is to invite you to learn about physics culture, history, and practices and to reflect on your growth as a physicist.

Each Exercise 0 will have a different prompt each week, but will be structured similarly with the expectation that you write several paragraphs in response to the prompt.

Some of the topics might challenge your prior experience with physics.

Course Activities

Reflection Activities (10%) – 3 forms of them **(turned in on D2L)**

3. Individual Homework Reflections (due with HW via the D2L Assignments page)

If you worked as a group (2-3 people):

- What was the most challenging part of this assignment, and how did you address it? This could be related to the content of the assignment, your group's dynamics, etc.
- Thinking about your role in the group, what did you contribute to your team's effort?
- Identify 1-2 specific positive interactions and 1-2 specific and actionable areas of improvement. Meanness and generalities are not constructive. Writing "no concerns, everything was great" is also not helpful. Think of ways your team could have been even better.

Course Activities

Midterm Projects (45%) -- 2 of them DUE Feb 27th and Apr 10th at 11:59pm **via Gradescope**

- Will look like the Homework but slightly beyond what you've already done
- **Can submit as group;** you can work together and with us
- Encouraged to use the textbook, notes, and other resources when solving the midterms and you can ask any of the teaching staff for help

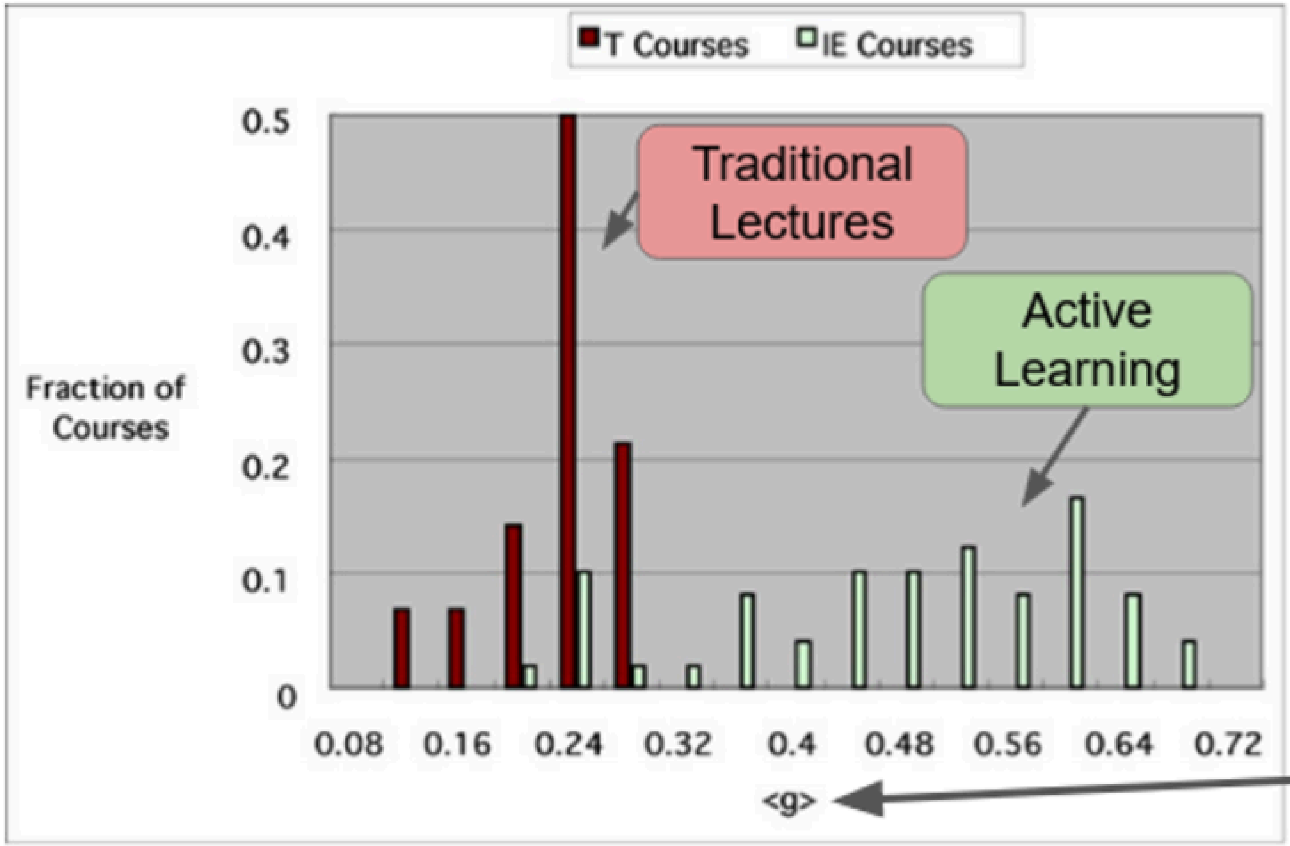
Final Project (30%) -- DUE Friday, Apr 27th and May 1st 11:59pm **via D2L**

- Analyze a physical system of your choice and prepare a narrative using Jupyter notebooks
- Can work in a group (up to 3 people) OR individually
- **Recorded presentations and review are part of this (during finals week)**

Extra Credit Opportunities

- Using iClickers during Class
  - Used in class to gauge your understanding of a topic or concept.
  - Will not be penalized for not knowing the correct answer.
  - Earn up to **1% extra credit to your overall homework grade**
- Attending the Department of Physics & Astronomy Seminar and/or Colloquia
  - Summarize the talk using at least 150-200 words and turn in your summary along with your homework assignment.
  - Earn up to **5 extra credit points on each homework assignment**
- Professional development (Midterm extra credit)
  - Develop a career or professional development plan. This plan will help you identify your goals, strengths, and areas for growth.
  - You can earn up to **5% extra credit on each midterm grade** by completing this plan and meeting with Danny to discuss it.

Measuring Student Learning



Larger is better

R. Hake, "…A six-thousand-student survey…" AJP **66**, 64-74 ('98).

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Generative AI Policy for PHY 321

- Next week, we will work together to develop a policy for our class.
- In preparing for that conversation, **please take at least 30 minutes to review the following link.**
  - https://openpraxis.org/articles/10.55982/openpraxis.16.4.777
- If you want more information, there are podcast episodes, videos, etc. posted on course website and D2L as well.
- Ask yourself:
  - What forms of AI use are acceptable to me? Why do I think that?
  - How would I demonstrate that I understand this material (i.e., your work is not just the product of AI)?
  - What forms of documentation should be used to indicate that I used AI tools?
  - When is it unacceptable to me to use AI in this class? Why do I think that?
  - What should happen if someone using AI in this class in a way that I find inappropriate?

Accommodations

If you have a university-documented learning difficulty or require other accommodations, please provide me with your VISA as soon as possible and speak with me about how I can assist you in your learning.

If you do not have a VISA but have been documented with a learning difficulty or other problems for which you may still require accommodation, please contact MSU's Resource Center for People with Disabilities (355-9642) in order to acquire current documentation.

If you need accommodations but do not yet have documentation, please contact me!

Questions?

Think, Pair, Share Activity

- *What are 1-2 things you are excited for this semester?*
- *What are 1-2 things you are nervous/anxious about this semester?*
- *What questions (if any) are still lingering for you?*

- *Think:* Take 4 minutes (I'll set a timer) to document your answers to these questions above
- *Pair:* Discuss your answers with the people around you (5 min).
- *Share:* Turn in your answers (you don't have to put your name on it but you can if you'd like me to know)

Writing & Discussion Activity

- What does learning physics mean to you? That is, what does it mean for you to **understand** physics.
  - Take 5 minutes and just write your ideas down.

Writing & Discussion Activity

- What does learning physics mean to you? That is, what does it mean for you to **understand** physics.
  - Take 5 minutes and just write your ideas down.
- Talk to your neighbors and develop a few ideas on what it means to **understand** physics
  - What topics/areas does it focus on? What kind of knowledge or ways of knowing or ways of doing physics are you focused on?
  - Do other people have different ways of thinking about what it means to understand physics?
- Post your ideas to https://raisemyhand.msucerl.org/

This week

- **Schedule**
  - Wednesday: Vectors and Newton's laws
  - Friday: Getting started with VS Code; programming activity
    - Prof. Brian O'Shea and Mihir Naik will teach on Friday
    - They will work Homework 1; Exercise 3 (if time)
- **What do you need to do?**
  - Read syllabus; respond to surveys; read notes for class
  - Generate questions for Wednesday's class!